

CyberTrust KSA

AI-Driven NCA Compliance Platform Developer Prototype & Technical Brief (Version 3.2)

1. Platform Brief for Technical Team

CyberTrust KSA v3.2 introduces Enterprise-Grade Continuous Monitoring and Automated Control Testing. The architecture now supports $24/7$ real-time detection of compliance drift, deep integration with 200+ security tools, and role-specific dashboards (Executive, BU Manager, IT/Security, Compliance Officer). This is built on top of the existing multi-framework engine (NCA + SABIC + Aramco).

2. Developer FAQ

Q: How does the Automated Control Testing work? A: A background microservice tests controls continuously (every 15 min to 24h) via API integrations with tools like Azure AD, Qualys, and AWS. Results are streamed to the frontend via WebSockets.

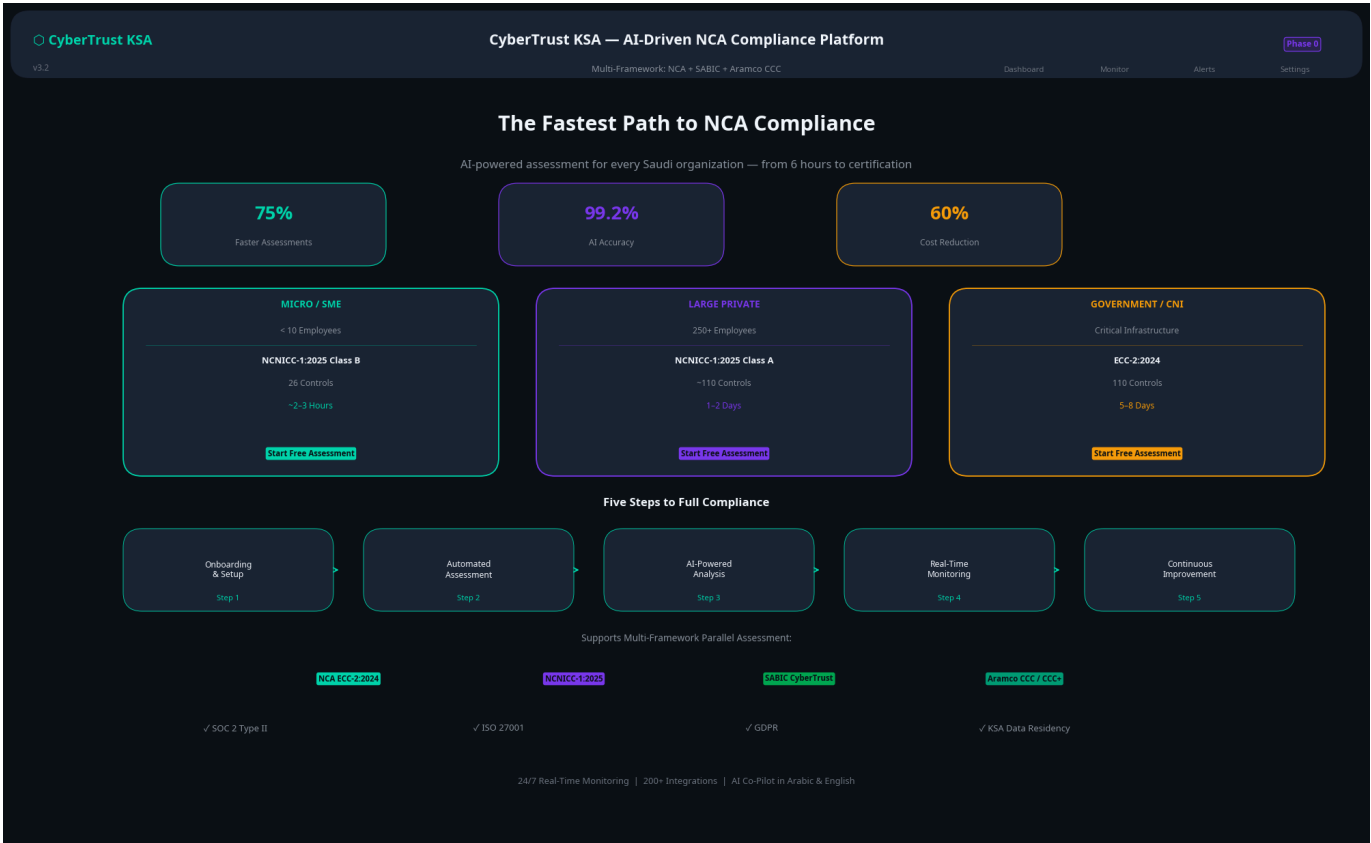
Q: How are the Role-Specific Dashboards implemented? A: Role is determined by JWT claims. The frontend router directs users to their specific view, and backend API endpoints (e.g., `/api/v1/dashboard/executive`) aggregate data tailored to that role's scope.

Q: How does Real-Time Monitoring detect changes? A: Through webhook listeners and API polling from the 200+ integrations. Any config change or access event triggers an evaluation against the active control frameworks, generating an alert if drift is detected.

3. Prototype Walkthrough & Scenarios (v3.2)

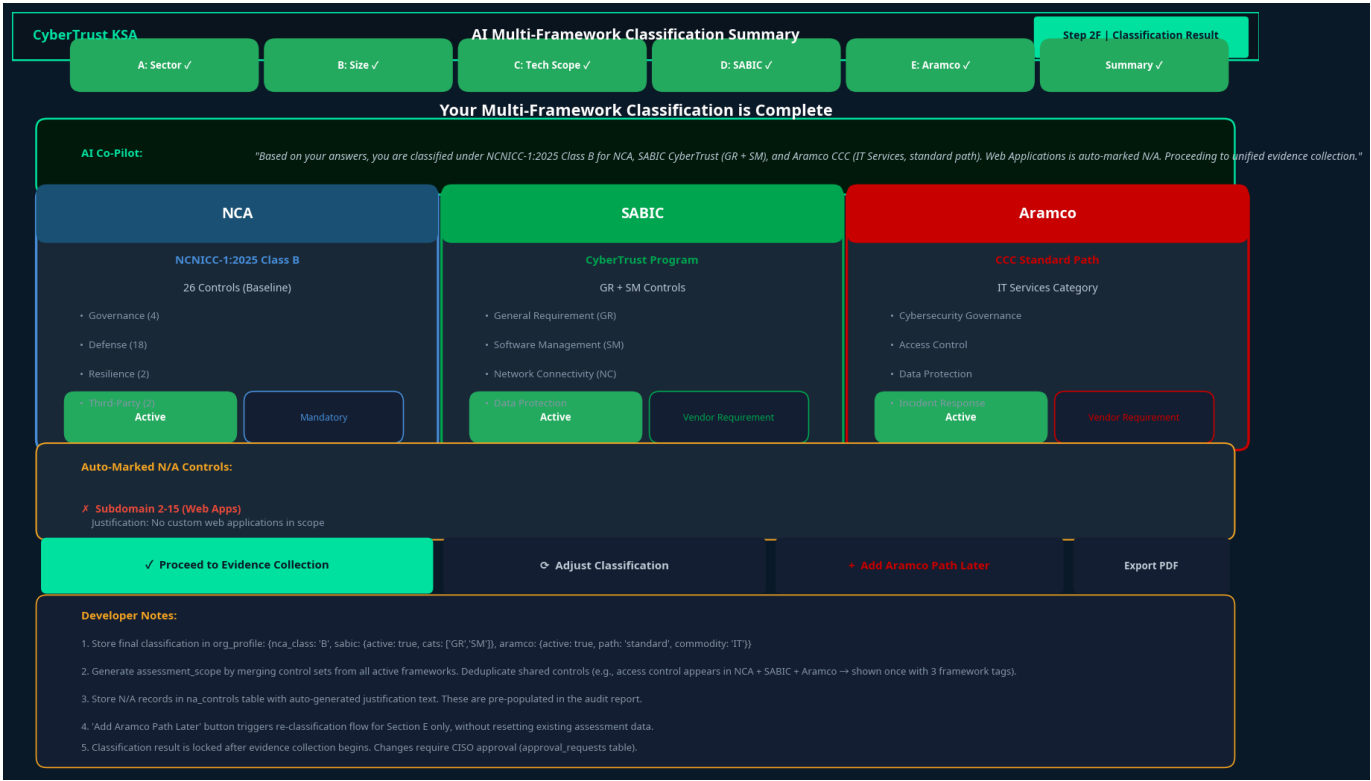
Phase 0: Landing Page (Updated)

Scenario: User lands on the platform. The new 5-step continuous process, ROI metrics, and security certifications are highlighted.



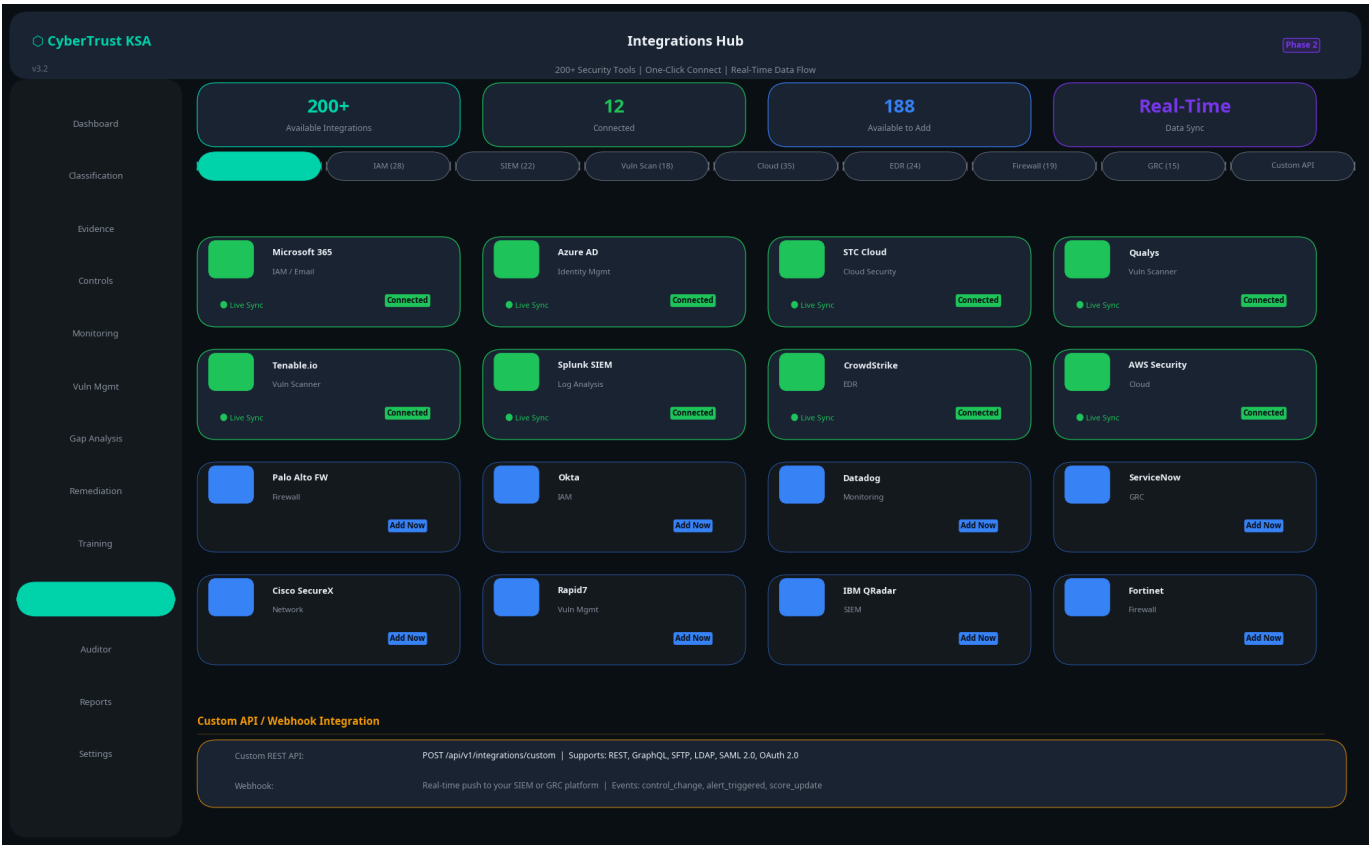
Phase 1: Smart Classification

Scenario: The AI Intake Questionnaire determines the exact control sets required based on Sector, Size, and Vendor targets (SABIC/Aramco).



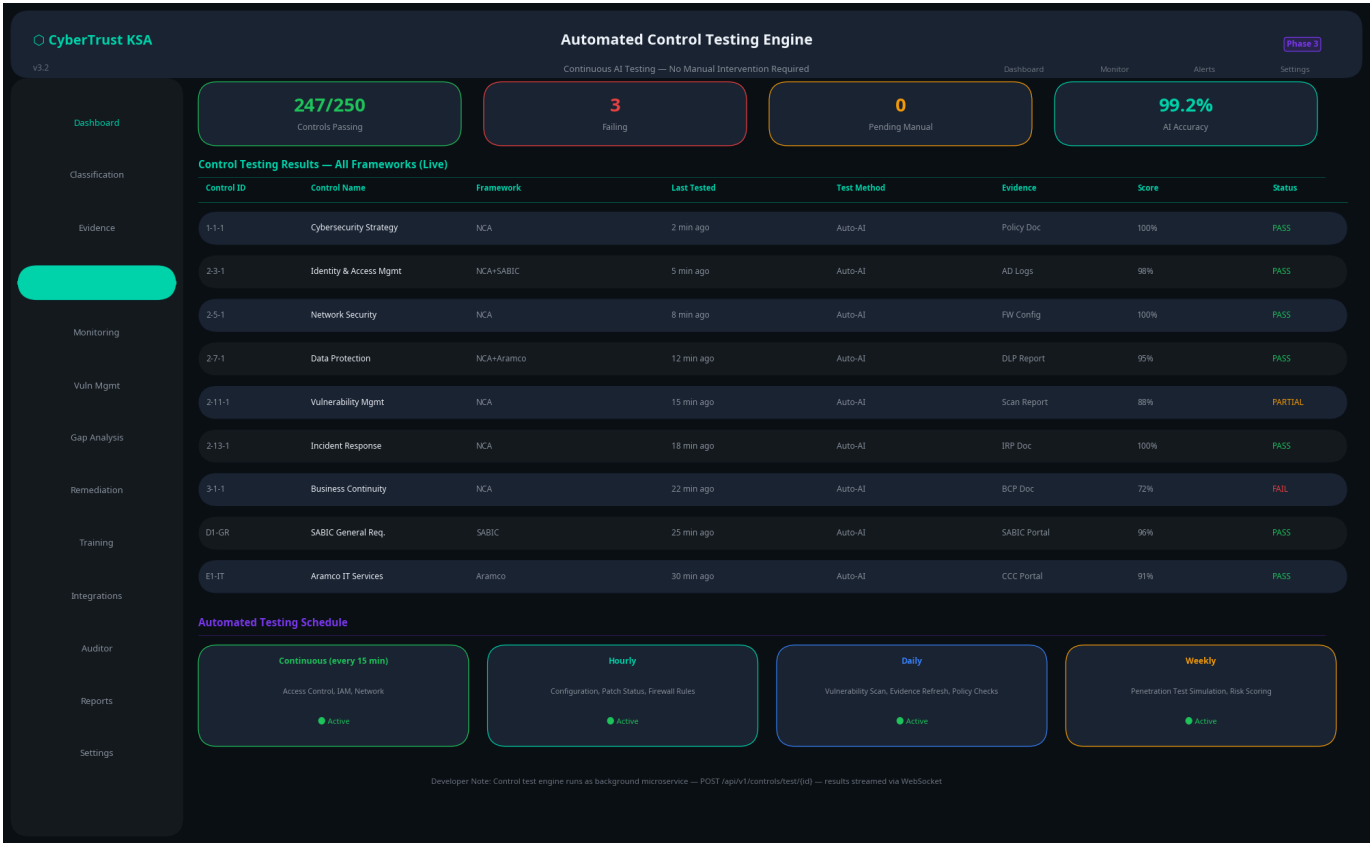
Phase 2: Integrations Hub (NEW)

Scenario: User connects their existing security stack. 200+ tools are available with one-click OAuth or API key setup, enabling real-time data flow.



Phase 3: Automated Control Testing (NEW)

Scenario: The AI engine continuously tests technical controls without manual intervention, showing live pass/fail status and evidence links.



Control Testing Results — All Frameworks (Live)

Control ID	Control Name	Framework	Last Tested	Test Method	Evidence	Score	Status
1-1-1	Cybersecurity Strategy	NCA	2 min ago	Auto-AI	Policy Doc	100%	PASS
2-3-1	Identity & Access Mgmt	NCA+SABIC	5 min ago	Auto-AI	AD Logs	98%	PASS
2-5-1	Network Security	NCA	8 min ago	Auto-AI	FW Config	100%	PASS
2-7-1	Data Protection	NCA+Aramco	12 min ago	Auto-AI	DLP Report	95%	PASS
2-11-1	Vulnerability Mgmt	NCA	15 min ago	Auto-AI	Scan Report	88%	PARTIAL
2-13-1	Incident Response	NCA	18 min ago	Auto-AI	IRP Doc	100%	PASS
3-1-1	Business Continuity	NCA	22 min ago	Auto-AI	BCP Doc	72%	FAIL
D1-GR	SABIC General Req.	SABIC	25 min ago	Auto-AI	SABIC Portal	96%	PASS
E1-IT	Aramco IT Services	Aramco	30 min ago	Auto-AI	CCC Portal	91%	PASS

Automated Testing Schedule

Continuous (every 15 min)

Access Control, IAM, Network

Active

Hourly

Configuration, Patch Status, Firewall Rules

Active

Daily

Vulnerability Scan, Evidence Refresh, Policy Checks

Active

Weekly

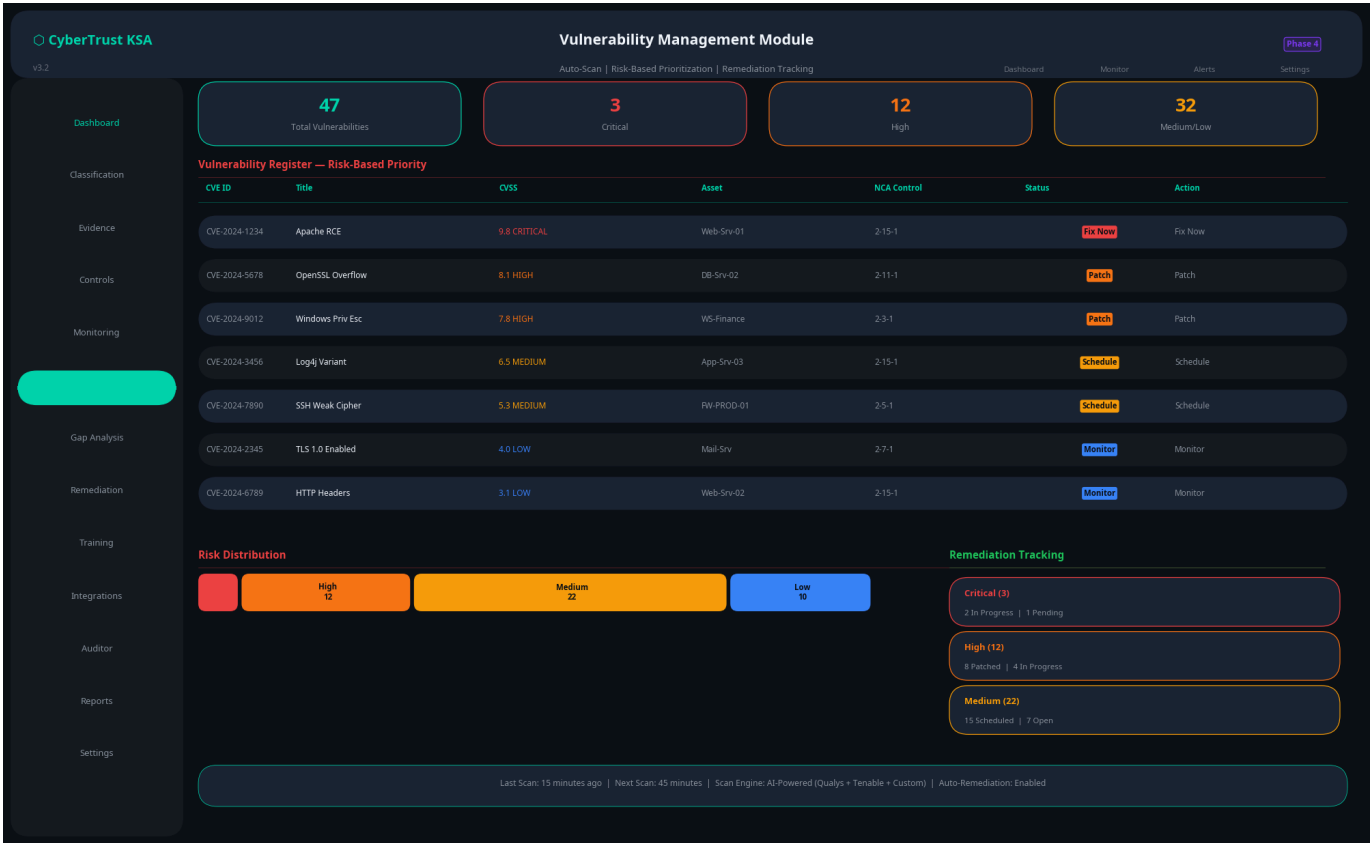
Penetration Test Simulation, Risk Scoring

Active

Developer Note: Control test engine runs as background microservice — POST /api/v1/controls/test/{id} — results streamed via WebSocket

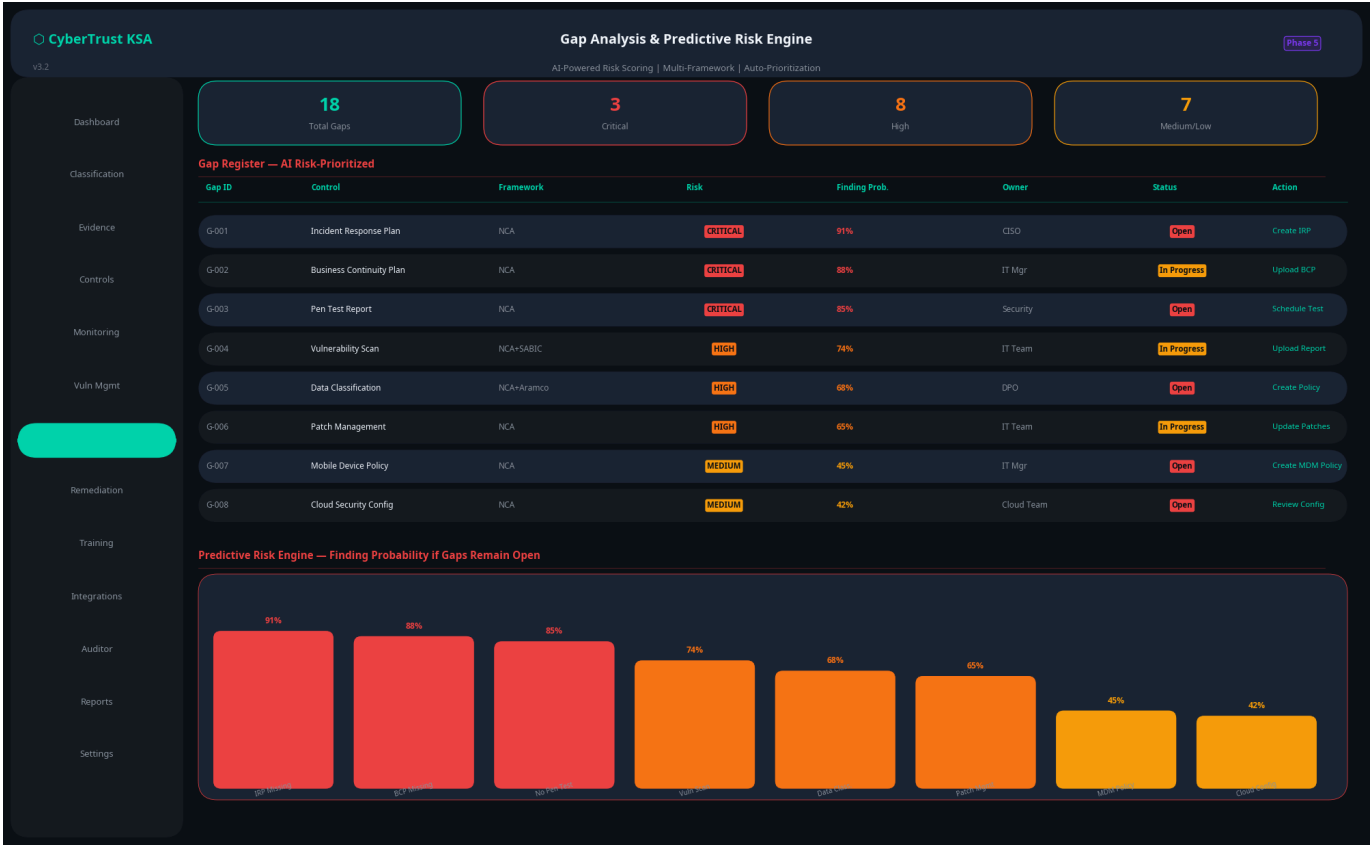
Phase 4: Vulnerability Management (NEW)

Scenario: Auto-scans pull CVEs, prioritize them by risk context, map them to NCA controls, and track remediation.



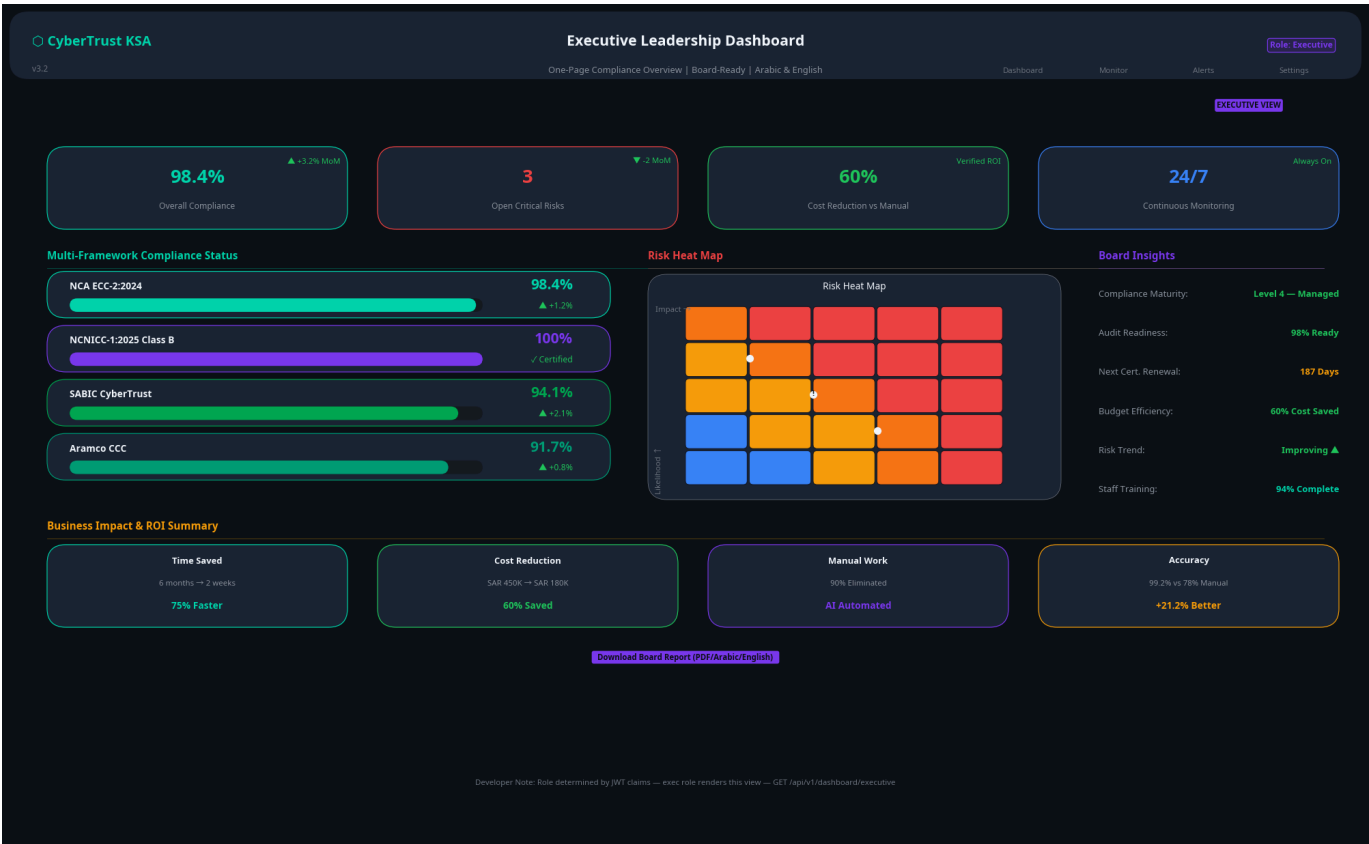
Phase 5: Gap Analysis & Predictive Risk (UPDATED)

Scenario: AI identifies gaps and calculates the probability of an auditor finding if left open.

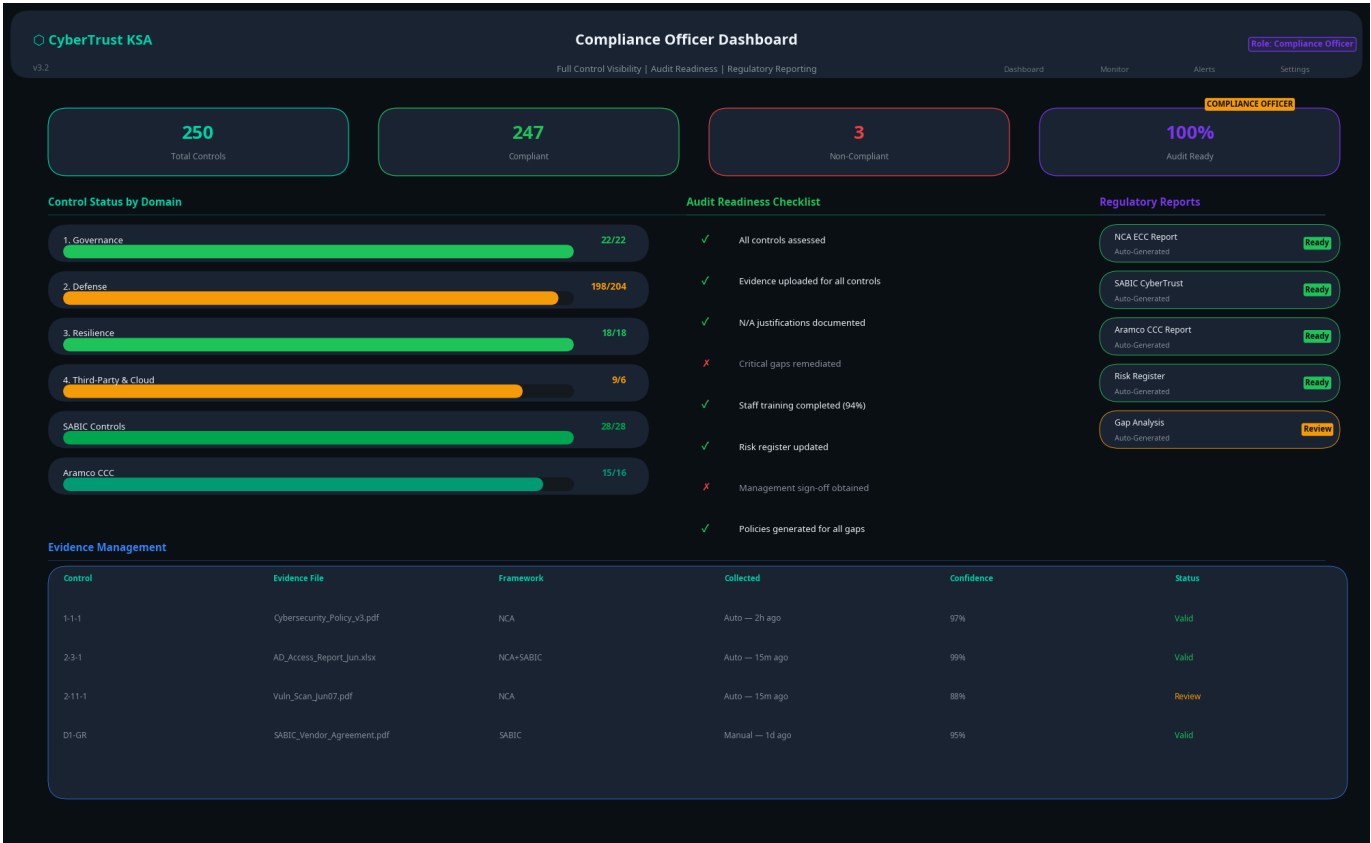


Phase 6: Role-Specific Dashboards (NEW)

A. Executive Leadership Dashboard Scenario: One-page overview with Risk Heat Map, ROI metrics, and board-ready reports.



B. Compliance Officer Dashboard Scenario: Full control visibility, audit readiness checklist, and regulatory reporting.



Control Status by Domain

Audit Readiness Checklist

Regulatory Reports

1. Governance

22/22

2. Defense

198/204

3. Resilience

18/18

4. Third-Party & Cloud

9/6

SABIC Controls

26/28

Aramco CCC

15/16

✓ All controls assessed

✓ Evidence uploaded for all controls

✓ N/A justifications documented

✗ Critical gaps remediated

✓ Staff training completed (94%)

✓ Risk register updated

✗ Management sign-off obtained

✓ Policies generated for all gaps

NCA ECC Report

Auto-Generated

Ready

SABIC CyberTrust

Auto-Generated

Ready

Aramco CCC Report

Auto-Generated

Ready

Risk Register

Auto-Generated

Ready

Gap Analysis

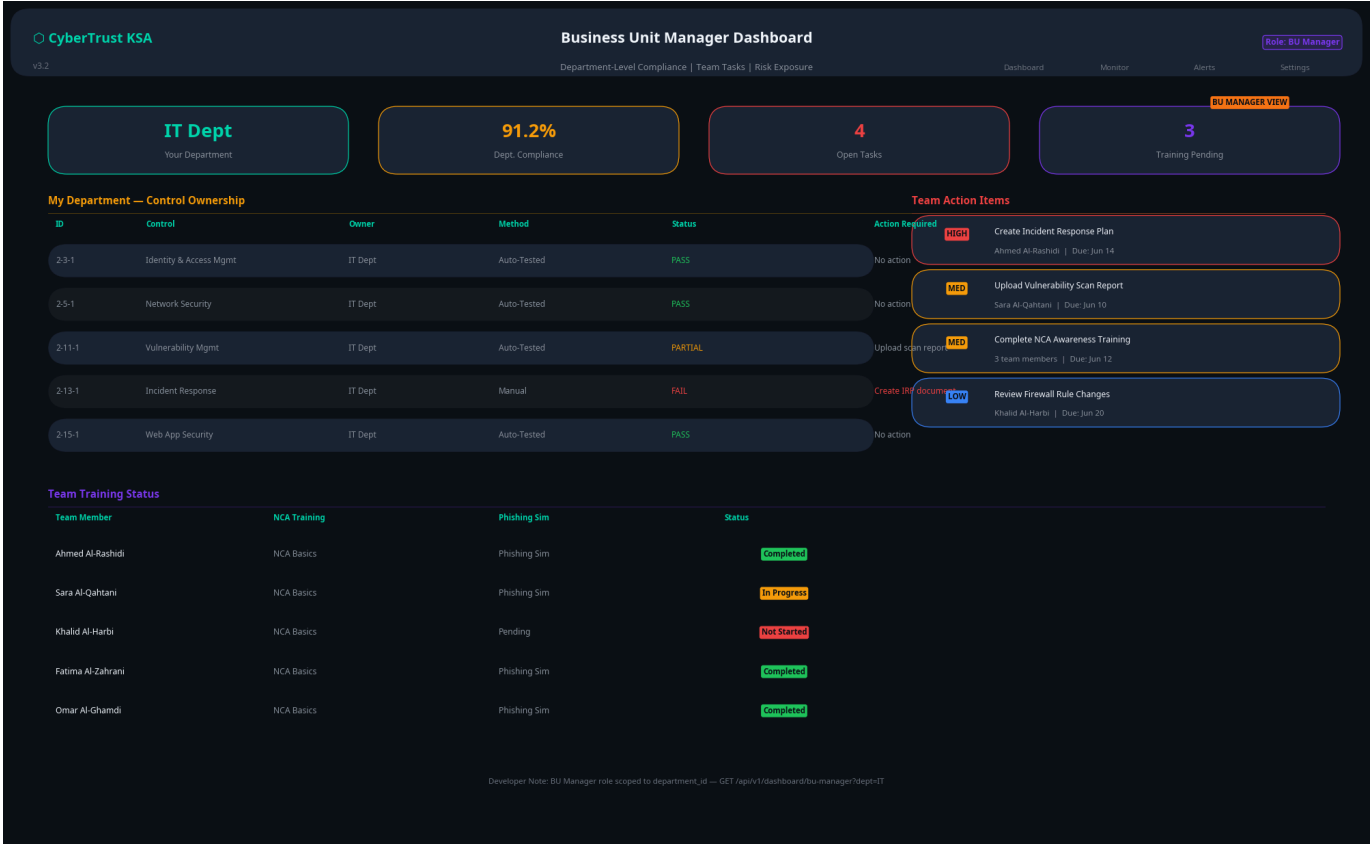
Auto-Generated

Review

Evidence Management

Control	Evidence File	Framework	Collected	Confidence	Status
1-1-1	Cybersecurity_Policy_v3.pdf	NCA	Auto — 2h ago	97%	Valid
2-3-1	AD_Access_Report_Jun.xlsx	NCA+SABIC	Auto — 15m ago	99%	Valid
2-11-1	Vuln_Scan_Jun07.pdf	NCA	Auto — 15m ago	88%	Review
DI-GR	SABIC_Vendor_Agreement.pdf	SABIC	Manual — 1d ago	95%	Valid

C. Business Unit Manager Dashboard Scenario: Department-level compliance, team task tracking, and staff training status.



My Department — Control Ownership

Team Action Items

ID	Control	Owner	Method	Status	Action Required
2-3-1	Identity & Access Mgmt	IT Dept.	Auto-Tested	PASS	No action
2-5-1	Network Security	IT Dept.	Auto-Tested	PASS	No action
2-11-1	Vulnerability Mgmt	IT Dept.	Auto-Tested	PARTIAL	Upload scan report
2-13-1	Incident Response	IT Dept.	Manual	FAIL	Create IR document
2-15-1	Web App Security	IT Dept.	Auto-Tested	PASS	No action

HIGH

Create Incident Response Plan

Ahmed Al-Rashidi | Due: Jun 14

MED

Upload Vulnerability Scan Report

Sara Al-Qahtani | Due: Jun 10

MED

Complete NCA Awareness Training

3 team members | Due: Jun 12

LOW

Review Firewall Rule Changes

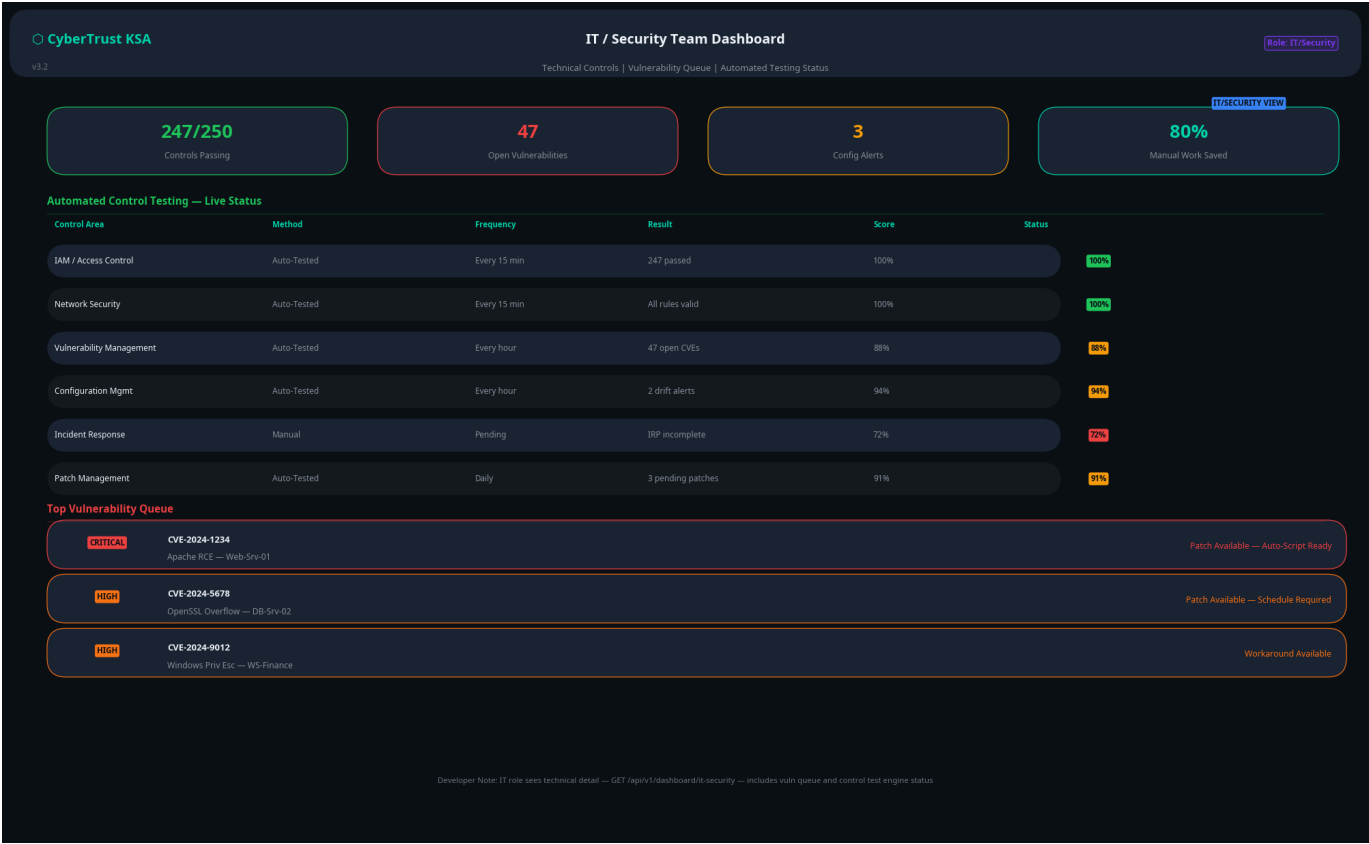
Khalid Al-Harbi | Due: Jun 20

Team Training Status

Team Member	NCA Training	Phishing Sim	Status
Ahmed Al-Rashidi	NCA Basics	Phishing Sim	Completed
Sara Al-Qahtani	NCA Basics	Phishing Sim	In Progress
Khalid Al-Harbi	NCA Basics	Pending	Not Started
Fatima Al-Zahrani	NCA Basics	Phishing Sim	Completed
Omar Al-Ghamdi	NCA Basics	Phishing Sim	Completed

Developer Note: BU Manager role scoped to department_id — GET /api/v1/dashboard/bu-manager?dept=IT

D. IT/Security Team Dashboard Scenario: Technical controls focus, vulnerability queue, and automated testing status.



Phase 7: Pre-Audit & Auditor Portal

Scenario: Virtual Auditor Simulator pre-tests readiness, then the real NCA Auditor logs in to review AI-analyzed evidence and submit the final report.

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Auditor Portal — Control Review & Findings

Step 10 | Auditor Review

My Cases

Control Review

Auditor Portal — NCA Approved Auditor

Case: Al-Rashidi Technology Solutions LLC | Framework: NCNICC-1:2025 Class B + SABIC CyberTrust | Readiness Score: 78%

Control-by-Control Review

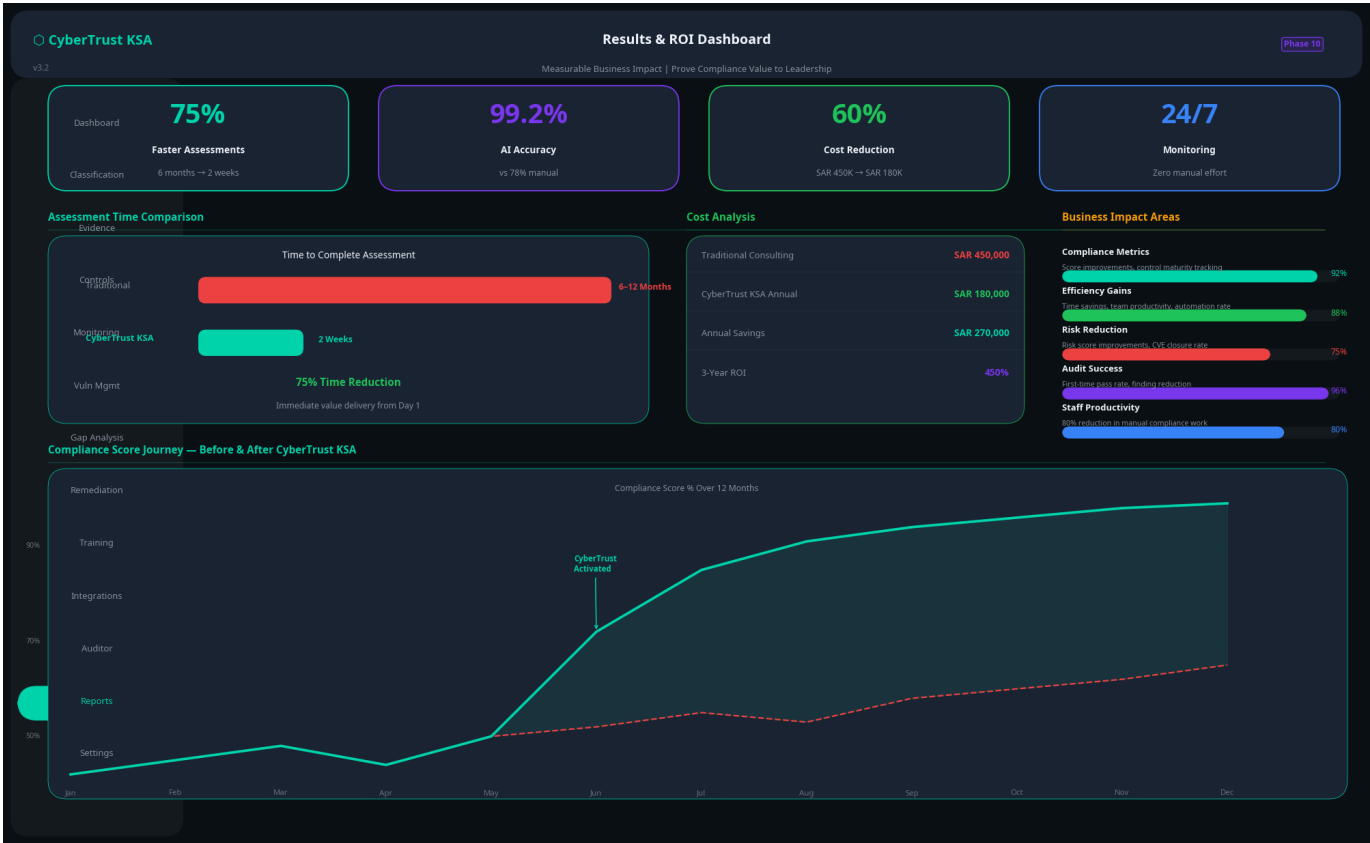
AI has pre-analyzed all evidence and provided a suggested verdict. Auditor can override with formal comments.

	Control	Name	Framework	AI Verdict	Evidence	Auditor Verdict	Comments
Documents	1-4	Roles & Responsibilities	NCA	Verified	Job Description...	Verified	—
	2-2	Identity & Access Mgmt	NCA+SABIC	Verified	Azure AD Logs	Verified	—
Findings	2-5	Network Security	NCA+Aramco	Conditional	Firewall Config	Conditional	Missing VLAN docs
	2-10	Vulnerability Mgmt	NCA+Aramco	Finding	No evidence	Finding	Critical gap — no ...
Submit Report	2-13	Incident Response	NCA+SABIC+Aramco	Finding	IRP v1.docx	Conditional	IRP missing RTO de...
	GR-1	SABIC Data Access Policy	SABIC	Verified	Policy v2.docx	Verified	—
	SM-3	Secure SDLC	SABIC	Conditional	SDLC Policy.doc...	Conditional	Needs code review ...
	2-15	Web App Security	NCA	N/A	Auto-excluded	N/A	Accepted justifica...

Submit Findings & Report →

Phase 8: Results & ROI Dashboard (NEW)

Scenario: Proves compliance value to leadership showing 75% faster assessments, 60% cost reduction, and business impact.



Phase 9: Security & Certifications (NEW)

Scenario: Highlights enterprise-grade security including SOC 2 Type II, ISO 27001, GDPR, and KSA Data Residency.

CyberTrust KSA

Enterprise Security & Certifications

Trust Center

v3.2

SOC 2 Type II | ISO 27001 | GDPR | KSA Data Residency

Your Data is Safe, Secure, and Always in Saudi Arabia

CyberTrust KSA meets the highest international security standards while ensuring full compliance with Saudi Arabia's data protection requirements

SOC 2 Type II

Security & Availability Controls

- Annual third-party audit
- All security controls verified
- Compliance documentation available
- Continuous monitoring validated

ISO 27001

Information Security Management

- International gold standard
- Comprehensive ISMS framework
- Risk management certified
- Annual surveillance audits

GDPR Compliant

Data Protection & Privacy

- Full GDPR compliance
- Data subject rights enforced
- Privacy by design architecture
- DPO appointed and active

KSA Data Residency

Saudi Data Law Compliance

- Data centers in Saudi Arabia
- No data transfer outside KSA
- Full PDPL compliance
- NCA-approved infrastructure

Security Architecture

End-to-End Encryption

AES-256 at rest, TLS 1.3 in transit

Zero-Trust Architecture

Every request authenticated and authorized

Multi-Factor Authentication

MFA enforced for all user accounts

Role-Based Access Control

Granular permissions per role and department

Immutable Audit Logs

Cryptographic hash chain for all evidence

Penetration Testing

Quarterly third-party pen tests

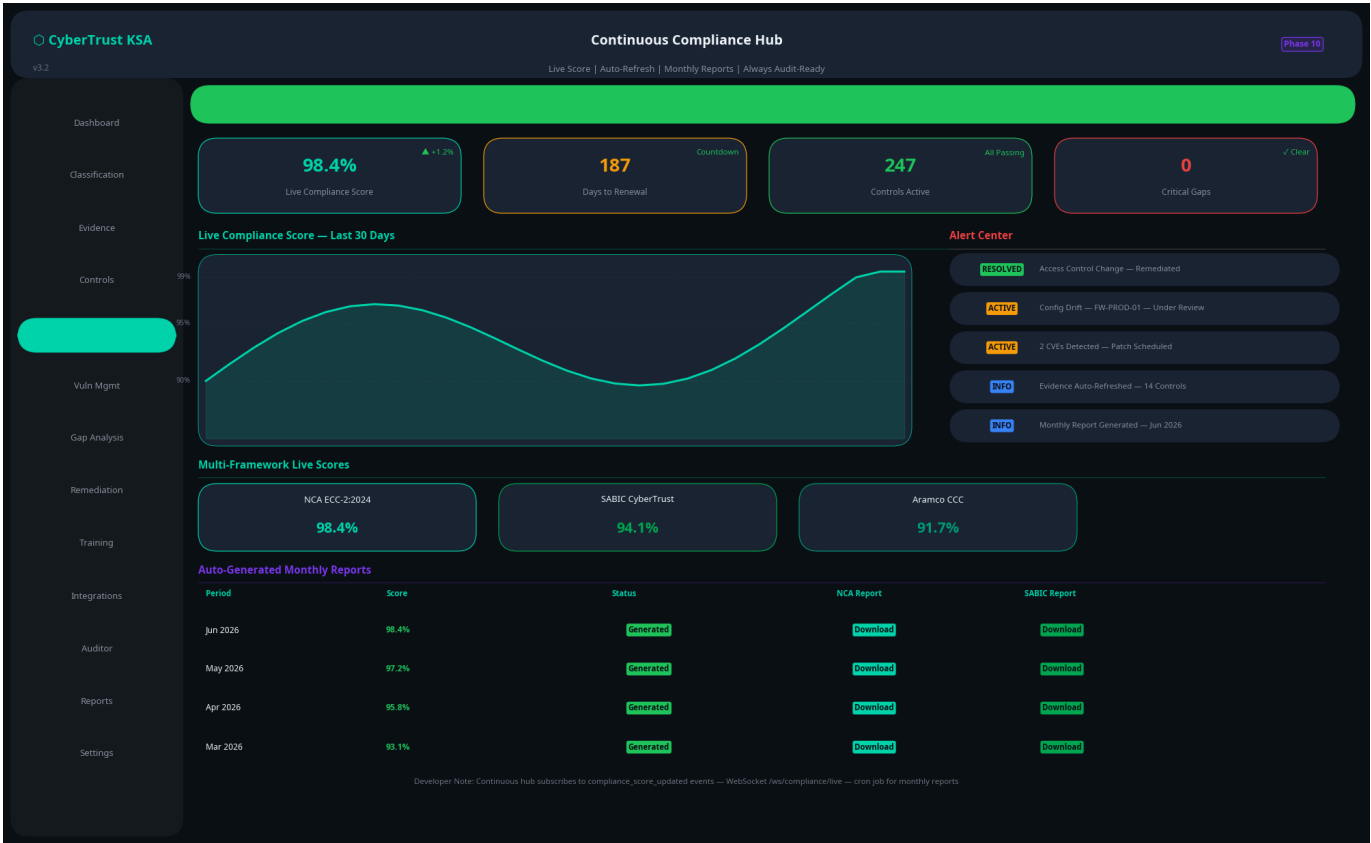
Infrastructure & Data Centers

Data Center	Location	Tier	Status
Primary DC	Riyadh, KSA	Tier IV	Active
Secondary DC	Jeddah, KSA	Tier III+	Hot Standby
Backup DC	NEOM, KSA	Tier III	Cold Standby
CDN Nodes	KSA + GCC	Edge	Active

All certifications and audit reports available upon request — security@cybertrust-ksa.sa

Phase 10: Continuous Compliance Hub & Real-Time Monitoring (UPDATED)

A. Continuous Compliance Hub Scenario: Live score updates daily, auto-generates monthly reports, and tracks renewal countdown.



B. Real-Time Monitoring Dashboard Scenario: $\frac{24}{7}$ detection of access control changes, config drift, and threats via a live event feed.

